CASHGUARD RISK

Know it’s safe before you pay.

An actuarial cash-safety check that tells small business owners the safest day to pay a bill, before cash runs short.

Team Extragorjoss · UiTM & UTeM · SDC Hackathon 2026 · Domain 2 · SDG 8.2 & 8.3

01 · THE PROBLEM

Owners decide what to pay by looking at today’s bank balance.

64 days

Average wait for Malaysian SMEs to get paid by customers

Experian, State of Credit 2025

48.7%

of Malaysia’s jobs come from MSMEs, so one cash crunch hits wages

DOSM, 2025

The balance hides what’s coming

Rent and payroll due next week. Customers who always pay late. Paying the wrong bill on the wrong day can leave no cash for wages.

02 · TARGET USERS

Micro and small businesses with no finance team

Owner

Cafés, workshops, shops. Decides every big payment and wants one clear answer: is it safe to pay?

Manager & staff

Capture invoices and request spending, only within the limits the owner sets.

Suppliers

Send invoices on WhatsApp and get told when they are approved and paid. No app to install.

03 · OUR SOLUTION

One number before every payment

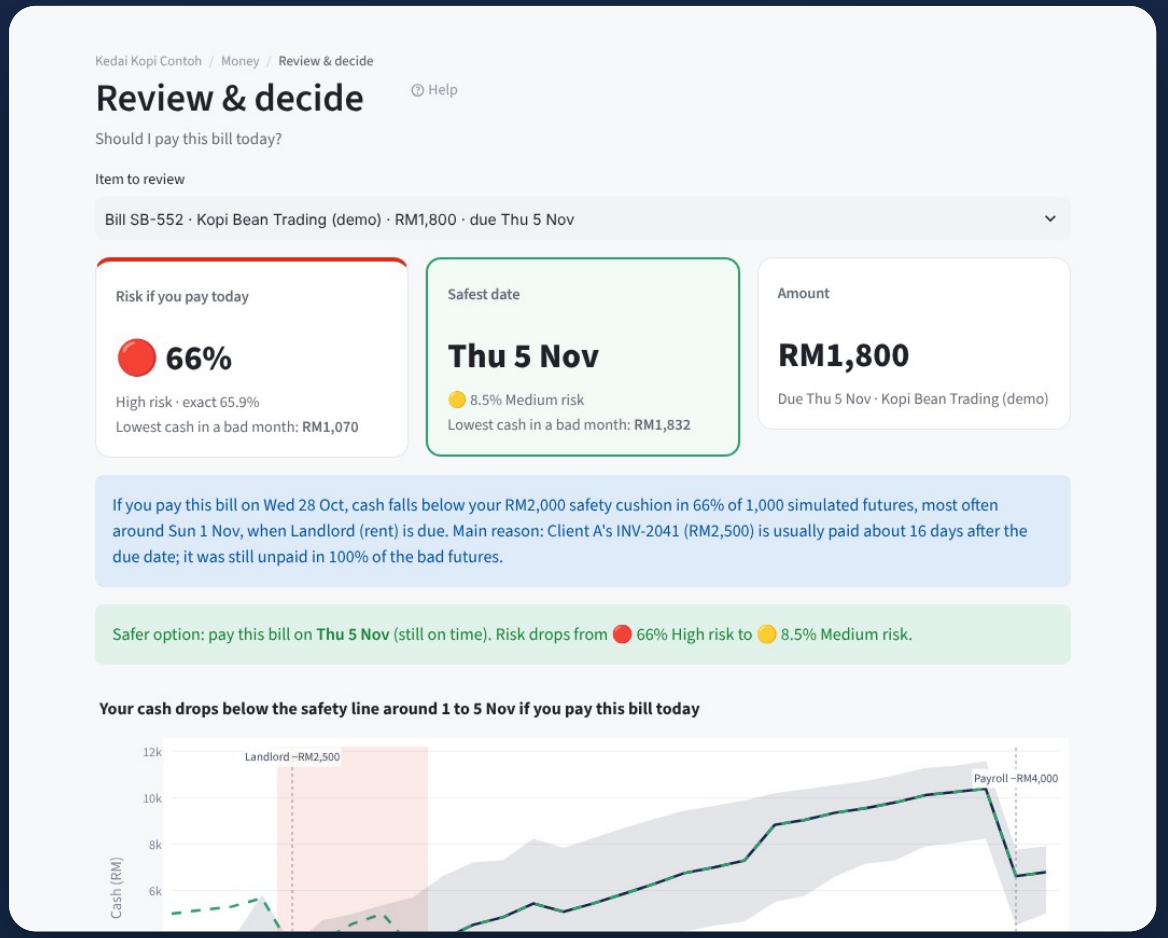
CashGuard Risk simulates the next 30 days 1,000 times and shows:

Chance of running short if you pay today (🟢 Low · 🟡 Medium · 🔴 High)

The safest pay date that is still on time

A plain reason: “rent is due on Sun 1 Nov”

The owner decides. AI never pays or approves anything.



04 · INNOVATION

Insurance-grade maths, made simple

Learns how each customer pays

Kaplan-Meier survival analysis plus Bühlmann credibility, the tools actuaries use.

A risk %, not a guess

1,000 Monte Carlo futures turn uncertainty into one honest number.

Safest date, never late

It only suggests dates up to the due date, so suppliers are always paid on time.

WhatsApp invoice bot

No invoice? Suppliers fill a WhatsApp form and the bot creates the PDF for them.

05 · HOW IT WORKS

From invoice to decision in five steps

1

Invoice arrives

WhatsApp photo, WhatsApp form or upload

2

Staff confirm

Checks for duplicates, totals and changed bank details

3

Risk engine

1,000 futures test each pay date

4

Owner decides

Pay now, schedule the safe date or hold

5

Supplier updated

WhatsApp status, saved in an audit log

Role-locked access: staff upload, managers decide within owner-set limits, the owner sees cash and risk. Every step is logged and can’t be edited.

06 · IMPACT & BENEFITS

77% fewer cash shortfalls, every bill still paid on time

14.4% → 3.3%

How often cash ran short: paying straight away vs. paying on CashGuard's date

Out-of-sample test on 50 synthetic bill situations, 300 fresh futures each

50 of 50 bills paid on time

4.4 days payment-timing error, vs 6.2 days naive

Owners protect wages and supplier trust **(SDG 8.2, 8.3)**

07 · FEASIBILITY

Already built, live and tested

Working today

Live app on Replit: cash-guard.replit.app

Open code on GitHub, Python + Streamlit

15 automated tests: maths, sign-in, limits, invoice form

Password sign-in with salted hashes and role-locked pages

Low cost: runs on a small server, no paid AI needed for the risk engine

Ready to switch on

AI invoice reading: GLM-OCR on RunPod, already coded

Live WhatsApp: Twilio / Meta webhook built

Production database: Supabase schema with row-level security

08 · IMPLEMENTATION

Roadmap from prototype to real cafés

PHASE 1 · NOW

Prototype

Risk engine, roles and limits, WhatsApp invoice form, live demo with synthetic data

PHASE 2 · NEXT ROUND

Connected

AI invoice reading on RunPod, live WhatsApp, Supabase database, “Ask CashGuard” chat

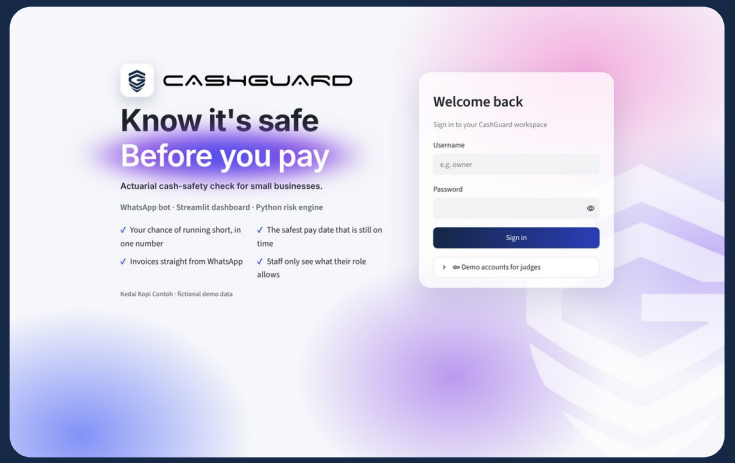
PHASE 3 · PILOT

Real businesses

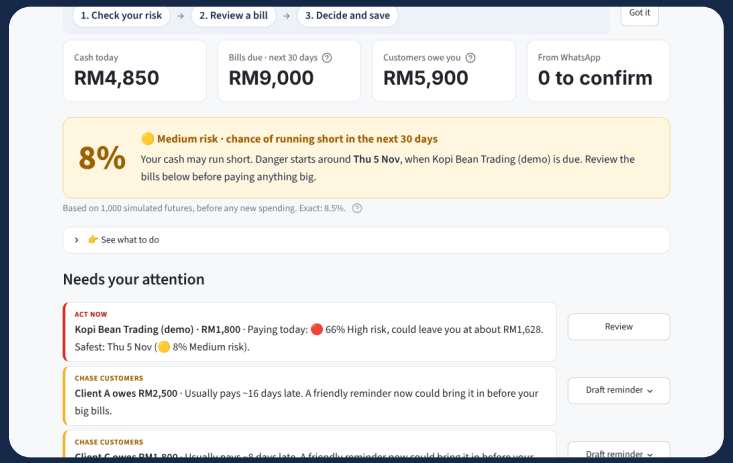
[Number] local SMEs, bank-statement import, check predictions on real payments

09 · VISUAL CONCEPT

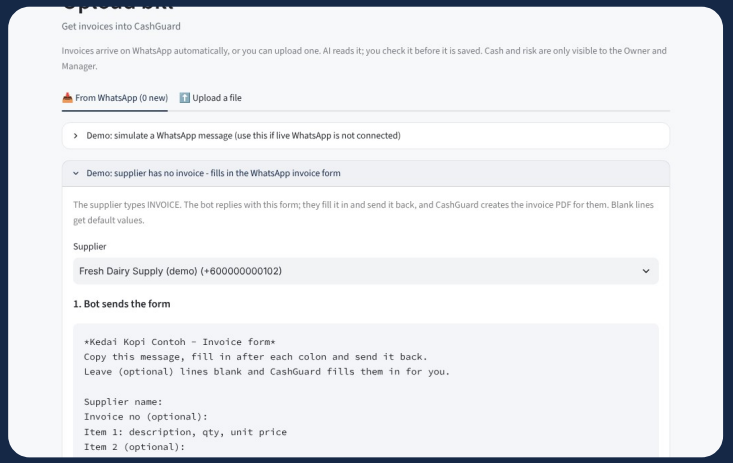
Clear enough for any shop owner



Sign in: each role sees only its pages



Dashboard: one risk %, what to do first



WhatsApp form: bot builds the invoice

Plain words, 🟢🟡🔴 risk labels and a Help button on every page that explains the maths.

10 · WHY WE SHOULD WIN

Real maths, real product, real proof

It works today

A live app judges can open and use, not slides about an idea.

Measured impact

77% fewer shortfalls in an out-of-sample test, with no late payments.

Responsible by design

AI never pays. Role limits, audit log, fraud checks on bank changes.

Actuarial + tech team

5 members from UiTM and UTeM combining actuarial science and software.

CASHGUARD

Try it now.

Live app: cash-guard.replit.app

Code: github.com/hecctarr/CashGuard

Sign in as owner / owner123 · manager / manager123 · staff / staff123

Thank you · Team Extragorjoss

